Program No-5

Basic Infrastructure Deployment

- **Tool**: Terraform
- **Program**:

1. Write a Terraform configuration file to provision a single EC2 instance on AWS.
2. Use Terraform commands (terraform init, terraform plan, terraform apply, terraform destroy) to manage the infrastructure.

Step 1: Create a Project Folder with name as terraform D drive and open that folder in VS code. Also install terraform extension in vscode

Step 2: Create the main.tf File

Create a file named main.tf and paste the following code:

```
provider "aws" {
   region = "us-west-2"
}
resource "aws_instance" "ec2_machine" {
  ami = "ami-07b0c09aab6e66ee9"
  instance_type = "t2.micro"
  //  count=4
  tags ={
    Name = "Terra EC2"
  }
}
```

- **provider "aws"** → tells Terraform - "We are using AWS cloud"
- **region = "us-west-2" -** Specifies *where* to create resources, This region = **Oregon (USA)** in Amazon Web Services
- **resource** → we are creating something
- **aws_instance** → type of resource (EC2 virtual machine)
- **ec2_machine** → name you give (local reference)
- **AMI = Amazon Machine Image**, It defines OS (Linux/Windows) and Pre-installed software
- **Instance Type -** Defines hardware CPU, RAM. **t2.micro**: Free tier eligible,Small machine
- **Count = 4** Terraform will create **4 EC2 instances**
- **Tags** are labels for AWS resources

Step 3: Initialize Terraform: execute following command on VS code terminal :

terraform init

This sets up Terraform in the folder and downloads necessary provider plugins.

Step 4: Preview the Plan

terraform plan

This shows what Terraform will create.

Step 5: Apply the Configuration

terraform apply

When prompted, type yes to confirm. Terraform will create the EC2 instance.

Step 6: Verify

Go to your **AWS Console → EC2 → Instances**, and you'll see the instance running.

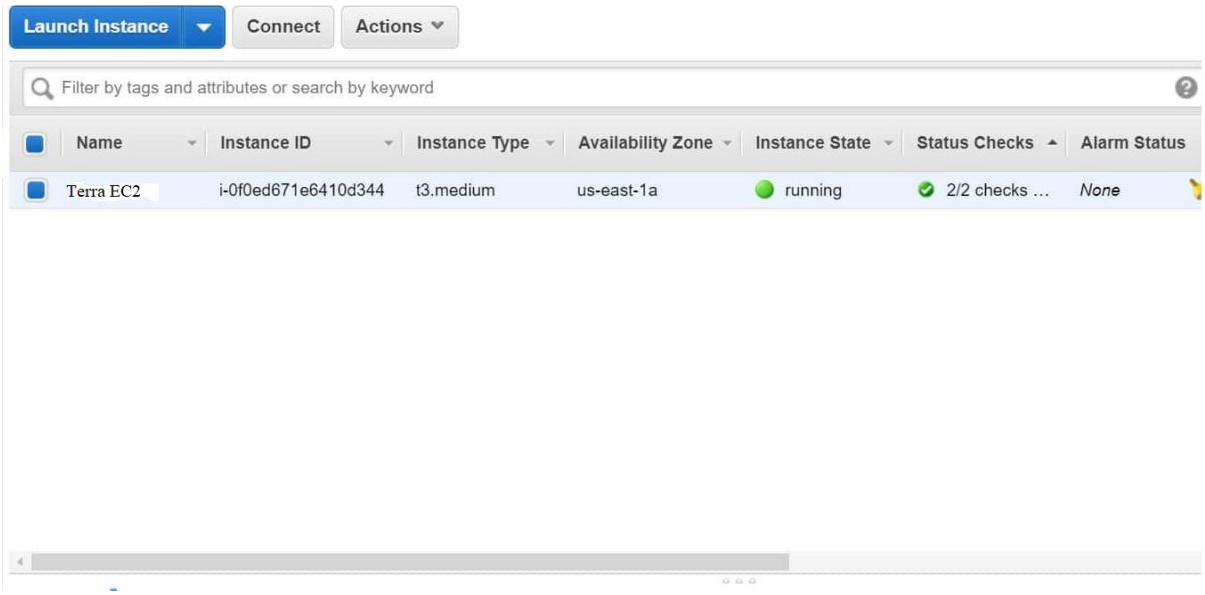
Step 7: Destroy When Done (to avoid charges)

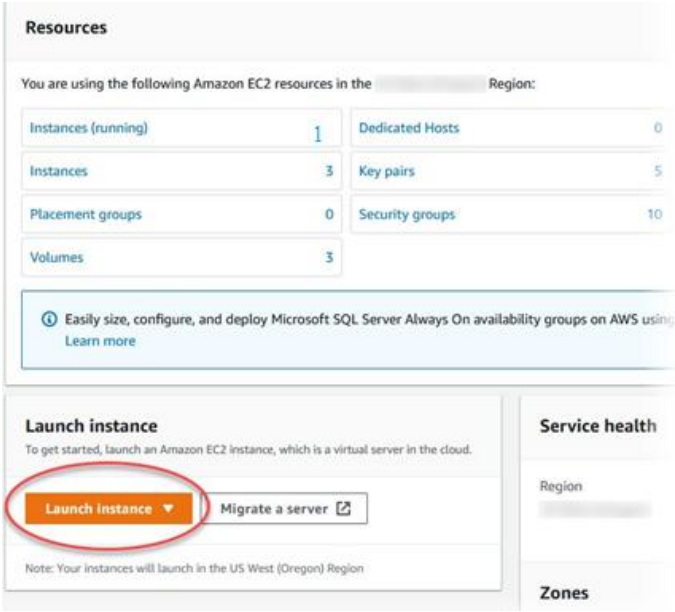
terraform destroy

Confirm with yes.

Output:

Instance Name: Terra EC2
Instance ID: i-0a1b2c3d4e5f67890
State: Running
Instance Type: t2.micro
Region: us-west-2
Public IP: 54.214.123.45




Resources
You are using the following Amazon EC2 resources in the Region:
Instances (running) 1
Dedicated Hosts 0
Instances 3
Key pairs 5
Placement groups 0
Security groups 10
Volumes 3
Easily size, configure, and deploy Microsoft SQL Server Always On availability groups on AWS usin
Learn more
Launch instance
To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.
Launch instance
Migrate a server
Note: Your instances will launch in the US West (Oregon) Region
Service health
Region
Zones